

Notes: Chen, Hong, Huang & Kubik (AER, 2004)
Does Fund Size Erode Mutual Fund Performance? The Role of Liquidity and Organization

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1 Big picture

- **Question.** Does fund size erode performance in active mutual fund management, and if so, is the mechanism liquidity, investor clientele, managerial agency, or organizational diseconomies?
- **Main empirical object.** The paper studies monthly performance of actively managed U.S. equity mutual funds from the CRSP Mutual Fund Database, primarily from 1962 to 1999, and relates performance to lagged fund size.
- **Main result.** Fund performance declines with lagged fund size. The result is present for both gross and net returns, and for several benchmarks: market-adjusted returns, CAPM-adjusted returns, Fama-French three-factor returns, and a four-factor model with momentum.
- **Economic magnitude.** In the baseline regressions, a two-standard-deviation increase in log fund size implies a performance decline of roughly 65–96 basis points per year, which is large relative to the average fund’s near-zero gross benchmark-adjusted performance.
- **Liquidity channel.** Scale hurts most among funds that have to invest in small and illiquid stocks. For small-cap growth funds, the size coefficient is strongly negative. For other funds, the interaction term largely offsets the negative effect.
- **Organization channel.** Family size does not hurt performance. Conditional on the size of the fund itself, the size of the fund family is positively related to performance. Scale is therefore not mechanically bad; how assets are organized matters.
- **Soft-information evidence.** Small funds and solo-managed funds appear better at investing in local stocks, a setting where soft information is important. This supports the idea that hierarchy and committee decision-making can limit the scalability of investment ideas.
- **Core takeaway.** Active fund management faces decreasing returns at the fund level when liquidity and organizational frictions prevent the fund from scaling its best ideas. But scale can be beneficial at the family level when separate funds preserve incentives and information-processing advantages.

2 Incremental contribution of the paper

- **From performance persistence to capacity constraints.** Earlier mutual fund work focused on whether managers have skill and whether performance persists. This paper adds the question of whether skill is scalable. It shows that fund performance depends not only on ability but also on the size of assets over which the ability is deployed.
- **Direct evidence on diseconomies of scale in active asset management.** The paper carefully documents a negative relation between fund size and future performance after controlling for style, past performance, fund age, expenses, turnover, flows, loads, and family size.
- **Separation of fund size from family size.** A key conceptual advance is to distinguish the assets managed by a specific fund from the total size of the fund family. The negative effect belongs to fund size, not family size. This distinction links mutual fund performance to organizational economics: scale can be good or bad depending on how decision rights and research are structured.
- **Mechanism: liquidity plus organization.** The paper goes beyond documenting a size-performance relation. It uses small-cap styles, local holdings, local stock returns, and management structure to argue that liquidity constraints interact with organizational diseconomies.
- **Evidence connecting asset management to the theory of the firm.** By using local investments as a proxy for soft information and team management as a proxy for organizational complexity, the paper brings ideas from hierarchy-cost theories into empirical asset management.
- **Investor implication.** Investors should not infer that a high-performing fund can absorb unlimited inflows. Past skill may become less valuable when the manager must deploy much more capital.

- **Theoretical implication.** The results are consistent with models such as Berk and Green in which flows chase performance until expected abnormal returns are competed away, but the paper contributes by showing empirically where and why decreasing returns arise.

3 Data and measurement

3.1 CRSP mutual fund sample

- The main data source is the CRSP Mutual Fund Database, covering mutual funds from 1962 to 1999.
- The authors restrict the sample to diversified U.S. equity mutual funds and exclude bond funds, international funds, and specialized sector funds.
- A fund must report total net assets under management and monthly returns.
- A fund must have at least one year of returns so that the authors can form lagged return and benchmark variables.
- Multiple share classes are cleaned to avoid treating share-class observations as separate funds.
- The baseline analysis excludes the smallest fund-size quintile because previous work warns that very small funds, especially funds under about \$15 million, may have upward-biased reported returns.

Interpretation: The baseline sample choice is designed to avoid spurious scale effects caused by data quality problems among very small funds. This also means that the paper is not trying to estimate whether the very smallest funds are inefficiently small. It focuses on whether medium-to-large active funds face decreasing returns.

3.2 Fund size and family size

- **TNA** is total net assets under management of a fund.
- **LOGTNA** is the natural logarithm of TNA.
- **LOGFAMSIZE** is the natural logarithm of one plus the assets under management of other funds in the same mutual fund family.
- The distinction is central: fund size captures scale inside the decision-making unit, while family size captures scale of the broader organization.

Interpretation: If large family size improves performance while large fund size reduces performance, the problem is not simply that asset management organizations become too big. The problem is how much capital is assigned to a single fund and decision process.

3.3 Controls and fund characteristics

- **TURNOVER** measures fund trading intensity and helps distinguish active funds from closet indexers.
- **AGE** is years since fund organization.
- **EXPRATIO** is annual management fees and expenses divided by TNA.
- **TOTLOAD** is total front-end, deferred, and rear-end load charges as a percentage of new investments.
- **FLOW** is percentage new fund flow over the previous year.
- **LAGFUNDRET** is cumulative buy-and-hold fund return over the previous 12 months.

Interpretation: These variables address the concern that fund size proxies for maturity, passiveness, high fees, past success, investor flows, or other fund characteristics rather than scale itself.

3.4 Return outcomes and performance benchmarks

The paper uses both gross and net fund returns.

- **Gross returns** are before fees and expenses.
- **Net returns** are after deducting fees and expenses.

Four performance benchmarks are used:

1. market-adjusted returns;
2. CAPM beta-adjusted returns;
3. Fama-French three-factor adjusted returns;
4. a four-factor model that adds momentum.

Interpretation: The benchmark adjustment is important because small funds may hold different types of stocks than large funds. A raw return comparison could confuse size with exposure to small, value, or momentum factors.

3.5 CDA Spectrum holdings data and local stocks

- CRSP does not contain fund-level holdings in individual stocks, so the authors use the CDA Spectrum Database for holdings analyses.
- CDA provides quarterly stock positions and becomes useful in the early 1980s.
- Local stocks are defined relative to the location of the fund headquarters and company headquarters, using geographic regions.
- The local-stock tests use the idea that local investing is more likely to rely on soft information, such as direct knowledge of firms and management.

Interpretation: The holdings data allow the paper to go beyond fund returns. The local-stock analysis asks whether small funds and different organizational structures actually invest differently in information-intensive opportunities.

3.6 Management structure

- The paper constructs a management-structure variable based on whether a fund is run by one manager or multiple managers.
- Multi-manager or committee-managed funds are interpreted as more hierarchical or coordination-intensive.
- The management-structure tests examine whether committee management affects local holdings, local investment performance, and overall fund performance.

Interpretation: This variable is an imperfect but useful proxy for organizational frictions. If soft-information investments require quick judgment and strong incentives, committee management may reduce performance in those settings.

4 Empirical specifications and model logic

4.1 Performance benchmark models

The paper adjusts fund returns using increasingly rich asset-pricing benchmarks. The CAPM regression is:

$$R_{i,t} = \alpha_i + \beta_{i,1}VWRF_t + \varepsilon_{i,t}. \quad (1)$$

The Fama-French three-factor model is:

$$R_{i,t} = \alpha_i + \beta_{i,1}VWRF_t + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + \varepsilon_{i,t}. \quad (2)$$

The four-factor model adds momentum:

$$R_{i,t} = \alpha_i + \beta_{i,1}VWRF_t + \beta_{i,2}SMB_t + \beta_{i,3}HML_t + \beta_{i,4}MOM12_t + \varepsilon_{i,t}. \quad (3)$$

Interpretation: These benchmark models remove systematic style exposures. The remaining adjusted return is interpreted as fund performance net of common factor exposures.

4.2 Baseline Fama-MacBeth performance regression

The main regression is:

$$FUNDRET_{i,t} = \mu + \phi LOGTNA_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t}, \quad (4)$$

where $FUNDRET_{i,t}$ is a benchmark-adjusted gross or net fund return and $X_{i,t-1}$ includes family size, turnover, age, expense ratio, total loads, fund flows, and lagged returns.

Interpretation: The coefficient ϕ measures whether larger funds underperform smaller funds after adjusting for style and observable characteristics. A negative ϕ is evidence of fund-level diseconomies of scale.

4.3 Style interaction specification for liquidity

To test whether the effect is strongest where liquidity matters, the paper estimates:

$$FUNDRET_{i,t} = \mu + \phi_1 LOGTNA_{i,t-1} + \phi_2 I_{style,i} + \phi_3 LOGTNA_{i,t-1} I_{style,i} + \gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (5)$$

The key style indicator is whether the fund is not a Small Cap Growth fund, or alternatively whether it is a Large Cap fund.

Interpretation: If scale hurts because of liquidity, then the negative size effect should be strongest for funds investing in small, illiquid stocks. A positive interaction for non-small-cap funds implies that the baseline negative size effect is concentrated among small-cap growth funds.

4.4 Fund family size specification

The paper also asks whether scale is bad at the family level:

$$FUNDRET_{i,t} = \mu + \phi LOGTNA_{i,t-1} + \psi LOGFAMSIZE_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (6)$$

Interpretation: If family size is positive while fund size is negative, scale is not inherently harmful. The same total assets can be organized across multiple funds in a way that preserves incentives, information-processing capacity, and manageable portfolio scope.

4.5 Local holdings and local performance specifications

The local-stock regressions are:

$$LocalHoldings_i = \alpha + \beta_1 SmallFund_i + \beta_2 SmallCapFund_i + \beta_3 (SmallFund_i \times SmallCapFund_i) + \delta' Controls_i + u_i, \quad (7)$$

$$LocalReturn_i = \alpha + \beta_1 SmallFund_i + \beta_2 SmallCapFund_i + \beta_3 (SmallFund_i \times SmallCapFund_i) + \delta' Controls_i + u_i. \quad (8)$$

Interpretation: The holdings regression asks whether small funds invest more in soft-information stocks. The returns regression asks whether they also perform better in those local investments. The interaction term is important because liquidity and soft information should matter most for small-cap funds.

4.6 Management structure specifications

For holdings and returns of local stocks:

$$Y_i = \alpha + \beta MultiManager_i + \delta' Controls_i + u_i, \quad (9)$$

where Y_i is either local holdings or returns on local holdings.

For fund performance:

$$FUNDRET_{i,t} = \alpha + \beta MultiManager_i + \phi LOGTNA_{i,t-1} + \psi LOGFAMSIZE_{i,t-1} + \gamma' X_{i,t-1} + \varepsilon_{i,t}. \quad (10)$$

Interpretation: Multi-manager funds may have more resources, but they may also suffer from committee and hierarchy costs. The paper interprets weaker local investing and lower performance for multi-manager funds as evidence that organizational structure limits scalability.

5 Identification strategy and threats to interpretation

5.1 Step 1: Use cross-sectional variation in lagged fund size

The paper avoids a fund fixed-effects strategy because changes in fund size can be mechanically generated by recent performance. If lucky recent performance increases fund size and performance then reverts to the mean, a within-fund design could falsely attribute underperformance to size. The authors instead use cross-sectional variation in lagged size.

Interpretation: This choice is conservative in one sense: large funds likely became large because they were good funds. If anything, unobserved skill could bias against finding that size reduces performance.

5.2 Step 2: Remove style-driven performance differences

The paper adjusts returns using market, CAPM, Fama-French, and four-factor benchmarks. This addresses the concern that small and large funds hold different types of stocks.

Interpretation: If the negative size effect survives all benchmark adjustments, it is less likely to be merely a style effect.

5.3 Step 3: Control for observable fund characteristics

The regressions control for family size, turnover, age, expenses, loads, flows, and lagged performance.

Interpretation: These controls help separate fund size from passiveness, maturity, cost structure, past success, and investor flow dynamics.

5.4 Step 4: Exclude the smallest funds

The main sample excludes fund size quintile one to avoid upward-biased returns of very small funds and data-quality problems.

Interpretation: This reduces the concern that the results are driven by anomalous tiny funds rather than true diseconomies of scale among active funds.

5.5 Step 5: Test mechanism through small-cap heterogeneity

The liquidity hypothesis predicts that size should matter most for small-cap funds. The paper therefore tests whether size effects are stronger among Small Cap Growth funds and weaker among Large Cap or non-small-cap funds.

Interpretation: This is the paper’s strongest mechanism test. It links scale effects to the liquidity of the assets the fund must trade.

5.6 Step 6: Separate fund scale from organizational scale

The family-size tests ask whether having a large organization is harmful. The result that family size improves performance suggests that scale can be beneficial when organized across separate funds.

Interpretation: This shifts the interpretation from “big organizations are bad” to “large pools of capital inside a single fund can be bad when ideas and trading capacity do not scale.”

5.7 Step 7: Use holdings, geography, and management structure for organizational evidence

The local-stock and management-structure tests are used to explore whether hierarchy costs and soft-information frictions explain why large funds do not simply hire more managers and scale up the number of ideas.

Interpretation: These tests are not a randomized experiment. They are best read as mechanism evidence consistent with organizational diseconomies.

5.8 Remaining limitations

- Fund size is not randomly assigned.
- Some unobserved fund quality or strategy differences may remain.
- The local-stock and management-structure analyses use cross-sectional holdings data for a shorter sample window.
- The organization results are exploratory; multi-manager status is an imperfect proxy for hierarchy costs.
- The paper cannot fully observe internal incentives, idea-generation processes, or trading costs at the order level.

Interpretation: The evidence is strongest for the reduced-form fact that fund size predicts lower performance and for the liquidity mechanism. The organizational channel is plausible and supported by several patterns but is less cleanly identified.

6 Tables and figures

6.1 Table 1 - Summary Statistics

Read:

- Panel A reports fund characteristics across size quintiles. Average TNA rises sharply across quintiles, from very small funds to funds with average assets above one billion dollars.
- Turnover, age, expense ratios, loads, and flows vary with size, motivating controls in the regressions.
- Panel D shows that smaller funds have higher raw market-adjusted returns than larger funds, both before and after expenses.

Interpretation: Table 1 motivates the whole paper. It shows a negative unconditional relation between fund size and returns, but also shows that fund size is correlated with many fund characteristics. This is why the paper needs benchmark-adjusted regressions.

Economic message: The table establishes the scale-performance puzzle and justifies the richer regression design.

TABLE 1—SUMMARY STATISTICS								
Panel A: Time-series averages of cross-sectional averages and standard deviations								
	Mutual fund size quintile					All funds	Quintiles 2-5	
	1	2	3	4	5			
Number of funds	148.2	147.8	147.8	147.8	147.3	741.4	590.6	
TNA (\$ million)	4.7 [3.2]	22.2 [7.2]	60.6 [16.6]	165.4 [54.8]	1164.7 [1797.1]	282.5 [925.8]	352.3 [1022.7]	
LOGTNA (\$ million)	1.09 [1.01]	2.94 [0.34]	3.96 [0.27]	4.94 [0.33]	6.45 [0.75]	3.87 [1.92]	4.57 [1.38]	
LOGFAMSIZE (\$ million)	3.70 [2.98]	4.52 [2.89]	5.29 [2.72]	5.91 [2.56]	7.00 [2.16]	5.28 [2.92]	5.68 [2.75]	
TURNOVER (% per year)	42.07 [83.03]	55.68 [74.00]	59.09 [68.60]	61.20 [64.28]	52.17 [54.56]	54.17 [71.84]	57.07 [67.21]	
AGE (years)	8.17 [8.54]	11.90 [10.73]	14.82 [12.85]	18.43 [14.38]	25.16 [15.06]	15.67 [13.96]	17.57 [14.33]	
EXPRATIO (% per year)	1.29 [1.11]	1.08 [0.59]	0.94 [0.46]	0.85 [0.37]	0.68 [0.31]	0.97 [0.68]	0.89 [0.48]	
TOTLOAD (%)	3.41 [3.32]	4.19 [3.32]	4.32 [3.34]	4.57 [3.39]	5.38 [2.88]	4.36 [3.36]	4.59 [3.32]	
FLOW (% per year)	30.79 [113.55]	30.66 [113.36]	26.97 [101.66]	21.27 [84.08]	24.67 [59.04]	23.13 [102.64]	23.13 [96.60]	
Panel B: Time-series averages of (monthly) correlations between fund characteristics (using all funds)								
	TNA	LOGTNA	LOGFAMSIZE	TURNOVER	AGE	EXPRATIO	TOTLOAD	FLOW
TNA	1.00	0.56	0.24	-0.05	0.27	-0.19	0.10	-0.03
LOGTNA		1.00	0.40	0.06	0.44	-0.31	0.19	-0.07
LOGFAMSIZE			1.00	0.08	0.08	-0.19	0.25	-0.01
TURNOVER				1.00	0.01	0.17	0.05	-0.01
AGE					1.00	-0.13	0.19	-0.18
EXPRATIO						1.00	-0.05	0.08
TOTLOAD							1.00	-0.04
FLOW								1.00
Panel C: Time-series averages of (monthly) correlations between fund characteristics (excluding smallest 20 percent of funds)								
	TNA	LOGTNA	LOGFAMSIZE	TURNOVER	AGE	EXPRATIO	TOTLOAD	FLOW
TNA	1.00	0.66	0.23	-0.08	0.24	-0.24	0.09	-0.03
LOGTNA		1.00	0.35	-0.05	0.37	-0.36	0.13	-0.07
LOGFAMSIZE			1.00	0.07	0.03	-0.17	0.22	-0.01
TURNOVER				1.00	-0.04	0.26	0.03	0.01
AGE					1.00	-0.18	0.17	-0.19
EXPRATIO						1.00	0.01	0.10
TOTLOAD							1.00	-0.05
FLOW								1.00
Panel D: Time-series averages of (monthly) cross-sectional averages of market-adjusted fund returns								
	Mutual fund size quintile					All funds	Quintiles 2-5	
	1	2	3	4	5			
FUNDRET (Gross)	0.09% [3.04%]	0.02% [2.64%]	0.03% [2.61%]	-0.06% [2.46%]	-0.06% [2.00%]	0.01% [2.62%]	-0.02% [2.48%]	
FUNDRET (Net)	-0.02% [3.04%]	-0.07% [2.64%]	-0.05% [2.61%]	-0.13% [2.46%]	-0.12% [2.00%]	-0.08% [2.62%]	-0.09% [2.48%]	

Notes: This table reports summary statistics for the funds in our sample. "Number of funds" is the number of mutual funds that meet our selection criteria for being an active mutual fund in each month. TNA is the total net assets under management in millions of dollars. LOGTNA is the logarithm of TNA. LOGFAMSIZE is the logarithm of one plus the assets under management of the other funds in the family that the fund belongs to (excluding the asset base of the fund itself). TURNOVER is fund turnover, defined as the minimum of aggregate purchases and sales of securities divided by the average TNA over the calendar year. AGE is the number of years since the establishment of the fund. EXPRATIO is the total annual management fees and expenses divided by year-end TNA. TOTLOAD is the total front-end, deferred, and rear-end charges as a percentage of new investments. FLOW

TABLE 2—SUMMARY STATISTICS FOR PERFORMANCE BENCHMARKS

Panel A: Summary statistics of the factors

Factor	Mean return	SD of return	Cross-correlations			
			VWRF	SMB	HML	MOM12
VWRF	0.58%	4.37%	1.00	0.32	-0.39	-0.02
SMB	0.17%	2.90%		1.00	-0.16	-0.30
HML	0.34%	2.63%			1.00	-0.15
MOM12	0.96%	3.88%				1.00

Panel B: Loadings calculated using the CAPM

Portfolio	CAPM				VWRF
	Alpha				
1 (small)	0.04%				0.87
2	-0.02%				0.91
3	-0.01%				0.93
4	-0.09%				0.92
5 (large)	-0.07%				0.91

Panel C: Loadings calculated using the 3-Factor model and the 4-Factor model

Portfolio	3-Factor model				4-Factor model				MOM12
	Alpha	VWRF	SMB	HML	Alpha	VWRF	SMB	HML	
1 (small)	0.01%	0.82	0.29	0.03	-0.02%	0.82	0.30	0.04	0.03
2	-0.03%	0.84	0.27	-0.01	-0.09%	0.85	0.30	0.00	0.05
3	0.01%	0.87	0.22	-0.05	-0.06%	0.87	0.25	-0.03	0.06
4	-0.06%	0.87	0.18	-0.05	-0.13%	0.87	0.20	-0.04	0.05
5 (large)	-0.05%	0.88	0.08	-0.06	-0.10%	0.88	0.10	-0.05	0.05

Notes: This table reports the loadings of the five (equal-weighted) TNA-sorted fund portfolios on various factors. Panel (A) reports the summary statistics for the factors. VWRF is the return on the CRSP value-weighted stock index in excess of the one-month Treasury rate. SMB is the return on a portfolio of small stocks minus large stocks. HML is the return on a portfolio long high book-to-market stocks and short low book-to-market stocks. MOM12 is the return on a portfolio long stocks that are past-12-month winners and short those that are past-12-month losers. Panel (B) reports the loadings calculated using the CAPM. Panel (C) reports the loadings calculated using the Fama-French (1993) 3-Factor model and this model augmented with the momentum factor (4-Factor model). The sample period is from January 1963 to December 1990.

6.2 Table 2 - Summary Statistics for Performance Benchmarks

Read:

- Panel A summarizes the benchmark factors: market excess returns, SMB, HML, and momentum.
- Panels B and C show the factor loadings of size-sorted mutual fund portfolios.
- Smaller fund portfolios load more on SMB and HML, while large funds load somewhat more on momentum.

Interpretation: Table 2 shows that fund-size portfolios are not identical in factor exposures. Without adjustment, a size-performance comparison might confuse manager skill with exposure to small-cap, value, or momentum factors.

Economic message: Benchmark adjustment is essential for making the fund-size coefficient interpretable as performance rather than style.

6.3 Table 3 - Regression of Fund Performance on Lagged Fund Size

Read:

- LOGTNA is negative across gross and net returns and across all four benchmarks.
- The coefficient is around -0.028 for market- and beta-adjusted gross returns and around -0.02 for three- and four-factor adjusted returns.
- LOGFAMSIZE is positive and statistically significant.
- Lagged fund returns are also positive, consistent with some persistence.

Interpretation: This is the central empirical table. The negative coefficient on LOGTNA means larger funds underperform after controlling for benchmark exposures and other fund characteristics. The positive family-size coefficient is already a clue that scale is not inherently harmful if organized differently.

Economic message: Fund-level assets appear to face decreasing returns, while family-level assets may provide resources or fixed-cost advantages.

TABLE 3—REGRESSION OF FUND PERFORMANCE ON LAGGED FUND SIZE

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
INTERCEPT	0.056 (0.96)	0.087 (1.63)	0.080 (1.28)	0.019 (0.30)	0.026 (0.44)	0.056 (1.05)	0.049 (0.79)	-0.011 (0.18)
LOGTNA _{<i>t-1</i>}	-0.028 (3.02)	-0.028 (3.04)	-0.023 (2.42)	-0.020 (2.15)	-0.025 (2.75)	-0.025 (2.76)	-0.020 (2.16)	-0.018 (1.89)
LOGFAMSIZE _{<i>t-1</i>}	0.007 (2.26)	0.007 (2.27)	0.007 (2.22)	0.007 (2.21)	0.007 (2.33)	0.007 (2.34)	0.007 (2.29)	0.007 (2.28)
TURNOVER _{<i>t-1</i>}	0.000 (0.85)	0.000 (0.86)	0.000 (0.85)	0.000 (0.83)	0.000 (0.77)	0.000 (0.78)	0.000 (0.77)	0.000 (0.75)
AGE _{<i>t-1</i>}	-0.001 (0.62)	-0.001 (0.62)	-0.001 (0.64)	-0.001 (0.63)	0.000 (0.52)	0.000 (0.53)	-0.001 (0.55)	-0.001 (0.54)
EXPRATIO _{<i>t-1</i>}	-0.004 (0.11)	-0.004 (0.09)	-0.007 (0.18)	-0.007 (0.18)	-0.039 (0.97)	-0.038 (0.95)	-0.041 (1.04)	-0.041 (1.05)
TOTLOAD _{<i>t-1</i>}	0.003 (1.26)	0.003 (1.25)	0.003 (1.26)	0.003 (1.29)	0.003 (1.21)	0.003 (1.20)	0.003 (1.21)	0.003 (1.25)
FLOW _{<i>t-1</i>}	0.000 (0.50)	0.000 (0.50)	0.000 (0.51)	0.000 (0.49)	0.000 (0.47)	0.000 (0.47)	0.000 (0.48)	0.000 (0.46)
LAGFUNDRET _{<i>t-1</i>}	0.029 (6.00)	0.028 (5.98)	0.028 (6.00)	0.029 (5.99)	0.029 (6.03)	0.029 (6.01)	0.029 (6.03)	0.029 (6.02)
No. of months	444	444	444	444	444	444	444	444

Notes: This table shows the Fama-MacBeth (1973) estimates of monthly fund returns regressed on fund characteristics lagged one month. The sample includes only funds that fall within fund size quintiles two to five. Fund returns are calculated before (gross) and after (net) deducting fees and expenses. These returns are adjusted using the market model, the CAPM, the Fama-French (1993) 3-Factor model, and the 4-Factor model. The dependent variable is *FUNDRET*. *LOGTNA* is the natural logarithm of *TNA*. *LOGFAMSIZE* is the natural logarithm of one plus the size of the family that the fund belongs to. *TURNOVER* is fund turnover and *AGE* is the number of years since the organization of the mutual fund. *EXPRATIO* is the total annual management fees and expenses divided by *TNA*. *TOTLOAD* is the total front-end, deferred, and rear-end charges

TABLE 4—REGRESSION OF FUND RETURNS ON LAGGED FUND SIZE, ROBUSTNESS CHECKS

Panel A: Sample includes all funds

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{<i>t-1</i>}	-0.023 (2.59)	-0.026 (2.94)	-0.019 (2.22)	-0.021 (2.48)	-0.020 (2.16)	-0.022 (2.49)	-0.015 (1.76)	-0.017 (2.02)
LOGFAMSIZE _{<i>t-1</i>}	0.006 (1.99)	0.006 (2.03)	0.006 (1.99)	0.006 (2.05)	0.006 (2.10)	0.006 (2.15)	0.006 (2.11)	0.006 (2.17)
No. of months	444	444	444	444	444	444	444	444

Panel B: Dependent variable is 12-month (non-overlapping) fund returns

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{<i>t-1</i>}	-0.436 (3.26)	-0.440 (3.27)	-0.345 (2.40)	-0.312 (2.19)	-0.402 (3.07)	-0.406 (3.07)	-0.312 (2.20)	-0.280 (1.99)
LOGFAMSIZE _{<i>t-1</i>}	0.088 (2.05)	0.089 (2.06)	0.088 (2.05)	0.088 (2.07)	0.090 (2.10)	0.091 (2.12)	0.090 (2.11)	0.090 (2.12)
No. of years	37	37	37	37	37	37	37	37

Panel C: Sample period is from 1963 to 1980

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{<i>t-1</i>}	-0.035 (2.41)	-0.034 (2.42)	-0.021 (1.51)	-0.019 (1.33)	-0.032 (2.22)	-0.032 (2.23)	-0.019 (1.52)	-0.016 (1.15)
LOGFAMSIZE _{<i>t-1</i>}	0.002 (0.49)	0.002 (0.50)	0.002 (0.45)	0.002 (0.44)	0.002 (0.56)	0.002 (0.58)	0.002 (0.53)	0.002 (0.52)
No. of months	216	216	216	216	216	216	216	216

Panel D: Sample period is from 1981 to 1999

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{<i>t-1</i>}	-0.021 (1.84)	-0.022 (1.86)	-0.024 (1.93)	-0.022 (1.71)	-0.019 (1.64)	-0.019 (1.65)	-0.022 (1.74)	-0.019 (1.53)
LOGFAMSIZE _{<i>t-1</i>}	0.012 (2.53)	0.012 (2.53)	0.012 (2.50)	0.012 (2.50)	0.012 (2.55)	0.012 (2.55)	0.012 (2.52)	0.012 (2.52)
No. of months	228	228	228	228	228	228	228	228

Notes: This table presents robustness checks of the regression specification in Table 3. In panel (A), the sample includes all funds. In panel (B), the dependent variable is 12-month fund returns and the regressions are non-overlapping. In panel (C), the sample consists only of observations from 1963 to 1980. In panel (D), the sample consists only of observations from 1981

6.4 Table 4 - Regression of Fund Returns on Lagged Fund Size, Robustness Checks

Read:

- Panel A includes all funds, including the smallest quintile. The negative LOGTNA effect remains.
- Panel B uses non-overlapping 12-month returns and still finds large negative size coefficients.
- Panels C and D split the sample into 1963–1980 and 1981–1999. The negative size effect appears in both periods, though significance varies with sample length.

Interpretation: Table 4 addresses three concerns: sample selection, monthly-return overlap, and subperiod instability. The result is not a fragile artifact of the baseline sample.

Economic message: The diseconomies of scale result is persistent across time and return definitions.

6.5 Table 5 - Effect of Fund Size on Performance by Fund Style

Read:

- Panel A focuses on Small Cap Growth funds versus other funds.
- LOGTNA is strongly negative for Small Cap Growth funds, around -0.055 to -0.063.
- The interaction with non-Small-Cap-Growth status is positive, around 0.040–0.044, partially offsetting the negative effect.
- Panel B shows a similar pattern for Large Cap funds: the interaction implies that size is much less damaging for funds that invest in more liquid large-cap stocks.

Interpretation: This table is the key liquidity test. If scale erodes performance because a large fund must trade in less liquid stocks and cannot deploy capital in its best ideas, the effect should be strongest for small-cap funds. That is exactly what the table shows.

Economic message: Liquidity, not merely marketing or passive behavior, is central to why size hurts active fund performance.

6.6 Table 6 - Effect of Family Size on Performance by Fund Style

Read:

- The table repeats the style-interaction logic but focuses on LOGFAMSIZE.
- Fund size remains negatively related to performance.
- Family size is not negatively related to performance; it is weakly positive or economically small.
- The family-size interaction with non-small-cap status does not generate the same adverse pattern as fund size.

Interpretation: Table 6 separates liquidity from organization. If simply being part of a large asset-management organization caused diseconomies, family size would also hurt. It does not.

Economic message: Scale can be beneficial when resources are spread across distinct funds. The problem is scaling a single portfolio and decision process.

Panel A: Small Cap Growth funds								
	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
INTERCEPT	0.205	0.257	0.320	0.263	0.175	0.226	0.289	0.231
	(1.27)	(1.53)	(2.44)	(1.98)	(1.08)	(1.34)	(2.20)	(1.75)
IND _(not SCG)	-0.250	-0.252	-0.260	-0.262	-0.249	-0.251	-0.259	-0.261
	(1.54)	(1.55)	(1.62)	(1.63)	(1.53)	(1.54)	(1.61)	(1.62)
LOGTNA _{t-1}	-0.058	-0.058	-0.063	-0.061	-0.055	-0.056	-0.060	-0.058
	(3.15)	(3.17)	(3.40)	(3.29)	(3.03)	(3.05)	(3.28)	(3.17)
LOGTNA _{t-1} * IND _(not SCG)	0.040	0.041	0.043	0.044	0.040	0.041	0.043	0.044
	(2.36)	(2.39)	(2.53)	(2.56)	(2.35)	(2.37)	(2.52)	(2.55)
LOGFAMSIZE _{t-1}	0.012	0.012	0.011	0.011	0.012	0.012	0.012	0.012
	(2.57)	(2.57)	(2.54)	(2.54)	(2.59)	(2.59)	(2.57)	(2.56)
TURNVER _{t-1}	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(1.62)	(1.61)	(1.62)	(1.61)	(1.57)	(1.56)	(1.57)	(1.57)
AGE _{t-1}	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
	(0.98)	(0.97)	(0.98)	(0.97)	(0.87)	(0.87)	(0.88)	(0.87)
EXPRATIO _{t-1}	-0.018	-0.018	-0.019	-0.019	-0.062	-0.062	-0.064	-0.063
	(0.44)	(0.43)	(0.46)	(0.44)	(1.49)	(1.48)	(1.52)	(1.51)
TOTLOAD _{t-1}	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
	(0.54)	(0.54)	(0.52)	(0.52)	(0.48)	(0.47)	(0.46)	(0.46)
FLOW _{t-1}	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.96)	(0.95)	(0.90)	(0.92)	(1.02)	(1.02)	(0.96)	(0.99)
FUNDRET _{t-1}	0.022	0.022	0.021	0.021	0.022	0.022	0.022	0.022
	(3.45)	(3.45)	(3.43)	(3.42)	(3.48)	(3.48)	(3.45)	(3.45)
No. of months	228	228	228	228	228	228	228	228

Panel B: Large Cap funds								
	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
INTERCEPT	0.016	0.068	0.128	0.068	-0.017	0.035	0.095	0.034
	(0.19)	(0.84)	(1.72)	(0.92)	(0.20)	(0.45)	(1.27)	(0.47)
IND _(LC)	-0.119	-0.122	-0.119	-0.120	-0.112	-0.114	-0.112	-0.113
	(1.31)	(1.33)	(1.33)	(1.34)	(1.23)	(1.25)	(1.24)	(1.26)
LOGTNA _{t-1}	-0.029	-0.030	-0.032	-0.029	-0.027	-0.027	-0.029	-0.027
	(2.22)	(2.24)	(2.45)	(2.25)	(2.02)	(2.05)	(2.26)	(2.05)
LOGTNA _{t-1} * IND _(LC)	0.031	0.031	0.031	0.031	0.030	0.030	0.030	0.030
	(2.29)	(2.32)	(2.34)	(2.36)	(2.23)	(2.26)	(2.28)	(2.30)
LOGFAMSIZE _{t-1}	0.011	0.011	0.011	0.011	0.011	0.011	0.011	0.011
	(2.54)	(2.54)	(2.51)	(2.51)	(2.58)	(2.58)	(2.55)	(2.55)
TURNVER _{t-1}	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(1.80)	(1.80)	(1.79)	(1.79)	(1.75)	(1.75)	(1.76)	(1.75)
AGE _{t-1}	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001	-0.001
	(1.26)	(1.25)	(1.26)	(1.25)	(1.17)	(1.17)	(1.17)	(1.16)
EXPRATIO _{t-1}	-0.020	-0.021	-0.020	-0.020	-0.065	-0.065	-0.066	-0.065
	(0.49)	(0.48)	(0.52)	(0.50)	(1.63)	(1.62)	(1.66)	(1.64)
TOTLOAD _{t-1}	0.001	0.001	0.001	0.001	0.001	0.001	0.001	0.001
	(0.35)	(0.35)	(0.34)	(0.35)	(0.30)	(0.30)	(0.29)	(0.29)
FLOW _{t-1}	0.000	0.000	0.000	0.000	0.000	0.000	0.000	0.000
	(0.84)	(0.82)	(0.77)	(0.79)	(0.89)	(0.88)	(0.83)	(0.85)
FUNDRET _{t-1}	0.020	0.020	0.020	0.020	0.020	0.020	0.020	0.020
	(3.10)	(3.09)	(3.07)	(3.07)	(3.13)	(3.12)	(3.10)	(3.09)
No. of months	228	228	228	228	228	228	228	228

Notes: This table reports the Fama-MacBeth (1973) estimates of monthly fund returns regressed on fund characteristics lagged one month. The sample includes only funds that fall within fund size quintiles two to five. Fund returns are calculated before (gross) and after (net) deducting fees and expenses. These fund returns are adjusted using the market model, the CAPM, the Fama-French (1993) 3-Factor model, and the 4-Factor model. In panel (A), the regression specification is the one in Table 3 but augmented with IND_(not SCG), which is a dummy variable that equals one if the self-reported fund style is Not Small Cap Growth and zero otherwise and this indicator variable interacted with LOGTNA. In panel (B), instead of IND_(not SCG), we augment the regression with IND_(LC), which is a dummy variable that equals one if the fund style is identified as a Large Cap fund. The sample is from January 1981 to December 1999. The t-statistics are adjusted for serial correlation using Newey-West (1987) lags of order three and are shown in parentheses.

Figure 1: Optimal production with differing ability across industries

TABLE 6—EFFECT OF FAMILY SIZE ON PERFORMANCE BY FUND STYLE

Panel A: Fund size quintiles two to five

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{t-1}	-0.051	-0.052	-0.056	-0.054	-0.048	-0.049	-0.053	-0.051
	(2.83)	(2.86)	(3.13)	(3.02)	(2.70)	(2.72)	(3.01)	(2.89)
LOGTNA _{t-1} *	0.033	0.034	0.036	0.037	0.033	0.033	0.036	0.036
	(1.94)	(1.98)	(2.12)	(2.15)	(1.93)	(1.96)	(2.10)	(2.14)
IND _(not SCG)	0.007	0.007	0.007	0.007	0.007	0.007	0.007	0.007
LOGFAMSIZE _{t-1}	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004
	(0.70)	(0.70)	(0.69)	(0.70)	(0.70)	(0.70)	(0.69)	(0.70)
LOGFAMSIZE _{t-1} *	0.004	0.004	0.004	0.004	0.004	0.004	0.004	0.004
	(0.41)	(0.41)	(0.41)	(0.40)	(0.42)	(0.42)	(0.42)	(0.41)
No. of months	228	228	228	228	228	228	228	228

Panel B: All fund size quintiles one to five

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
LOGTNA _{t-1}	-0.067	-0.070	-0.071	-0.072	-0.064	-0.067	-0.068	-0.069
	(3.67)	(3.83)	(3.87)	(3.91)	(3.51)	(3.67)	(3.71)	(3.76)
LOGTNA _{t-1} *	0.065	0.065	0.067	0.066	0.065	0.065	0.067	0.066
	(3.52)	(3.52)	(3.61)	(3.56)	(3.53)	(3.53)	(3.62)	(3.57)
IND _(not SCG)	0.012	0.012	0.012	0.012	0.012	0.012	0.012	0.012
	(1.13)	(1.13)	(1.11)	(1.11)	(1.15)	(1.15)	(1.13)	(1.13)
LOGFAMSIZE _{t-1} *	-0.003	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002	-0.002
	(0.23)	(0.21)	(0.23)	(0.21)	(0.23)	(0.21)	(0.22)	(0.21)
No. of months	228	228	228	228	228	228	228	228

Notes: This table reports the Fama-MacBeth (1973) estimates of monthly fund returns regressed on fund characteristics lagged one month. Fund returns are calculated before (gross) and after (net) deducting fees and expenses. These returns are adjusted using the market model, the CAPM, the Fama-French (1993) 3-Factor model, and the 4-Factor model. The regression specification is the one in Table 5 but augmented with IND_(not SCG), which is a dummy variable which equals one if the self-reported fund style is Not Small Cap Growth and zero otherwise, interacted with LOGFAMSIZE. Estimates of the intercept and the other independent variables are omitted for brevity. The other independent variables include TURNVER,

Figure 2: Model with no common managerial ability across industries (Panel A) and with managerial ability applicable across industries (Panel B)

6.7 Table 7 - Effect of Fund and Family Size on Investments in Local Stocks

Read:

- SMALLFUND predicts a larger portfolio share in local stocks.
- SMALLCAPFUND also predicts more local holdings.
- The interaction $\text{SMALLFUND} \times \text{SMALLCAPFUND}$ is positive and large.
- FAMILYSIZE has little effect.

Interpretation: Local stocks proxy for soft-information opportunities. The table suggests that small funds, especially small-cap funds, are more willing or better able to allocate capital to local soft-information investments.

Economic message: Small funds appear to exploit localized information more intensively than large funds.

6.8 Table 8 - Effect of Fund and Family Size on the Performance of Investments in Local Stocks

Read:

- SMALLFUND has a positive effect on returns to local investments.
- The interaction with SMALLCAPFUND is positive and statistically meaningful in some specifications.
- Family size again has little effect.

Interpretation: Table 8 adds performance to the holdings result. Small funds do not merely hold more local stocks; they appear to do better in local-stock investments, especially where soft information and illiquidity interact.

Economic message: The local-stock evidence supports the idea that large funds lose performance because their organizational structure does not scale soft information efficiently.

6.9 Table 9 - Effect of Management Structure on the Amount and Performance of Investments in Local Stocks

Read:

- MULTI-MANAGER is negative for both local holdings and local-stock returns.
- The results include fund size, family size, style, city, momentum, and book-to-market effects.
- The point estimates suggest that team-managed funds are less intensive and less successful in local soft-information investments.

Interpretation: Table 9 connects the mechanism to internal organization. Committee or multi-manager structures may be worse at processing and acting on soft information.

Economic message: Organization matters for scale: adding managers does not automatically solve the problem of deploying more capital.

TABLE 7—EFFECT OF FUND AND FAMILY SIZE ON INVESTMENTS IN LOCAL STOCKS

	(1)	(2)	(3)
<i>SMALLFUND</i>	0.0759 (9.49)	0.0580 (6.74)	0.0584 (6.79)
<i>SMALLCAPFUND</i>	0.0515 (6.28)	0.0340 (3.86)	0.0341 (3.88)
<i>SMALLFUND</i> * <i>SMALLCAPFUND</i>		0.1244 (5.58)	0.1229 (5.80)
<i>FAMILYSIZE</i>			0.0001 (1.01)
Momentum effects	Yes	Yes	Yes
Book-to-market effects	Yes	Yes	Yes
Fund city effects	Yes	Yes	Yes

Notes: This table shows the effect of fund and family size on mutual fund investments in local stocks. The dependent variable is the percentage of the value of a fund's portfolio invested in local stocks. *SMALLFUND* equals one if a fund's size is in the bottom 10 percent of the fund size distribution and zero otherwise. *SMALLCAPFUND* equals one if a fund's Daniel, Grinblatt, Titman, and Wermers (1997) small cap style score is in the bottom 10 percent across funds. *FAMILYSIZE* is the number of funds in the family that the fund belongs to. Momentum effects and book-to-market effects control for other differences in fund styles. Data are from the end of September 1997. Robust *t*-statistics are reported in parentheses.

(a) Table 7: Local holdings

TABLE 8—EFFECT OF FUND AND FAMILY SIZE ON THE PERFORMANCE OF INVESTMENTS IN LOCAL STOCKS

	(1)	(2)	(3)
<i>SMALLFUND</i>	0.0258 (1.83)	0.0143 (0.94)	0.0150 (0.99)
<i>SMALLCAPFUND</i>	0.1190 (8.21)	0.1078 (6.95)	0.1079 (6.96)
<i>SMALLFUND</i> * <i>SMALLCAPFUND</i>		0.0795 (2.02)	0.0776 (1.96)
<i>FAMILYSIZE</i>			0.0002 (1.02)
Momentum effects	Yes	Yes	Yes
Book to market effects	Yes	Yes	Yes
Fund city effects	Yes	Yes	Yes

Notes: This table shows the effect of fund and family size on the performance of mutual fund investments in local stocks. The dependent variable is the 4-quarter return to fund investments in local stocks. *SMALLFUND* equals one if a fund's size is in the bottom 10 percent of the fund size distribution and zero otherwise. *SMALLCAPFUND* equals one if a fund's Daniel, Grinblatt, Titman, and Wermers (1997) small cap style score is in the bottom 10 percent across funds. *FAMILYSIZE* is the number of funds in the family that the fund belongs to. Momentum effects and book-to-market effects control for other differences in fund styles. Data are from the end of September 1997. Robust *t*-statistics are reported in parentheses.

(b) Table 8: Local performance

TABLE 9—EFFECT OF MANAGEMENT STRUCTURE ON THE AMOUNT AND PERFORMANCE OF INVESTMENTS IN LOCAL STOCKS

	Holdings (1)	Returns (2)
<i>MULTI-MANAGER</i>	-0.0102 (-1.75)	-0.0190 (-1.73)
Fundsize effects	Yes	Yes
Family size effects	Yes	Yes
Momentum effects	Yes	Yes
Book-to-market effects	Yes	Yes
Style effects	Yes	Yes
Fund city effects	Yes	Yes

Notes: This table shows the effect of fund management structure on the holdings and performance of mutual fund investments in local stocks. The dependent variable in col-

(c) Table 9: Management structure and local investments

6.10 Table 10 - Effect of Fund Management Structure on Performance

Read:

- MULTI-MANAGER is negative across gross and net performance measures.
- The coefficient is about -0.040 monthly, or about -48 basis points annually.
- Fund size remains negative, and family size remains positive.

Interpretation: Table 10 extends the management-structure result from local holdings to overall performance. Committee-managed funds underperform even after controlling for fund size and other characteristics.

Economic message: The table strengthens the organizational interpretation: performance erosion is not only about the number of dollars under management; it is also about how investment decisions are made.

TABLE 10—EFFECT OF FUND MANAGEMENT STRUCTURE ON PERFORMANCE

	Gross fund returns				Net fund returns			
	Market-Adj	Beta-Adj	3-Factor	4-Factor	Market-Adj	Beta-Adj	3-Factor	4-Factor
<i>INTERCEPT</i>	-0.069 (0.66)	-0.018 (0.19)	0.039 (0.25)	-0.008 (0.05)	-0.092 (0.88)	-0.042 (0.44)	0.014 (0.09)	-0.032 (0.21)
<i>MULTI-MANAGER_t</i>	-0.040 (2.22)	-0.040 (2.23)	-0.041 (2.30)	-0.041 (2.30)	-0.040 (2.26)	-0.041 (2.27)	-0.042 (2.34)	-0.042 (2.34)
<i>LOGTNA_{t-1}</i>	-0.024 (2.34)	-0.023 (2.30)	-0.026 (1.98)	-0.025 (1.83)	-0.022 (2.19)	-0.022 (2.15)	-0.025 (1.86)	-0.023 (1.70)
<i>LOGFAMSIZE_{t-1}</i>	0.016 (2.38)	0.016 (2.38)	0.016 (2.38)	0.016 (2.38)	0.016 (2.40)	0.016 (2.39)	0.016 (2.40)	0.016 (2.39)
<i>TURNOVER_{t-1}</i>	0.000 (1.28)	0.000 (1.29)	0.000 (1.28)	0.000 (1.29)	0.000 (1.26)	0.000 (1.26)	0.000 (1.26)	0.000 (1.26)
<i>AGE_{t-1}</i>	-0.001 (0.64)	-0.001 (0.65)	-0.001 (0.64)	-0.001 (0.64)	-0.001 (0.60)	-0.001 (0.61)	-0.001 (0.59)	-0.001 (0.59)
<i>EXPRATIO_{t-1}</i>	-0.002 (0.03)	-0.002 (0.03)	-0.003 (0.04)	-0.003 (0.04)	-0.043 (0.54)	-0.043 (0.54)	-0.045 (0.55)	-0.045 (0.55)
<i>TOTLOAD_{t-1}</i>	-0.002 (0.28)	-0.002 (0.29)	-0.002 (0.30)	-0.002 (0.29)	-0.002 (0.36)	-0.002 (0.36)	-0.002 (0.37)	-0.002 (0.37)
<i>FLOW_{t-1}</i>	0.000 (1.64)	0.000 (1.68)	0.000 (1.68)	0.000 (1.69)	0.000 (1.64)	0.000 (1.68)	0.000 (1.68)	0.000 (1.69)
<i>FUNDRET_{t-1}</i>	0.030 (3.38)	0.030 (3.38)	0.029 (3.34)	0.029 (3.34)	0.031 (3.41)	0.030 (3.41)	0.030 (3.37)	0.030 (3.37)
No. of months	96	96	96	96	96	96	96	96

Notes: This table reports the Fama-MacBeth (1973) estimates of monthly fund returns regressed on fund characteristics lagged one month. The sample includes only funds that fall within fund size quintiles two to five. Fund returns are calculated before (gross) and after (net) deducting fees and expenses. These returns are adjusted using the market model, the CAPM, the Fama-French (1993) 3-Factor model, and the 4-Factor model. The regression specification is the one in Table 3 but augmented with *MULTI-MANAGER_t*, which is a dummy variable that equals one if the fund is managed by two or more individuals or by a team. The sample is from January 1995 to December 1999. The *t*-statistics are adjusted for serial correlation using

7 Interpretive synthesis

7.1 How the evidence fits together

The paper’s logic is cumulative:

1. Table 3 establishes that fund size predicts lower performance.
2. Table 4 shows that this result is robust.
3. Table 5 shows that the effect is concentrated where liquidity is most binding.
4. Table 6 shows that family scale is not harmful.
5. Tables 7–10 show that soft-information and organizational frictions help explain why large funds cannot simply hire more people and scale their best ideas.

Interpretation: The paper does not rely on a single regression. It builds a mechanism by moving from reduced-form performance to liquidity heterogeneity, organizational scale, portfolio holdings, local-stock performance, and management structure.

7.2 Why liquidity alone is not enough

Liquidity explains why a large fund cannot put all its money into its best small-stock ideas. But liquidity alone does not explain why a large fund cannot hire additional managers and generate more good ideas. The family-size and management-structure tests address this second step.

Interpretation: The paper’s most important conceptual point is that liquidity and organization are complements. Illiquidity creates the need for more ideas; hierarchy costs make idea generation and implementation hard to scale.

7.3 What investors should learn

- A skilled active manager may have limited capacity.
- Large inflows can dilute future performance even if past performance was genuine skill.
- Investors should distinguish fund size from fund family size.
- Funds that rely on small-cap or local information may be especially capacity constrained.

8 Short summary

- The paper studies whether active mutual fund performance declines with fund size.
- It uses CRSP mutual fund data from 1962 to 1999 and CDA holdings data for portfolio composition tests.
- Returns are measured gross and net of fees and adjusted using market, CAPM, Fama-French, and four-factor benchmarks.
- The baseline Fama-MacBeth regressions show that larger funds underperform smaller funds.
- The economic magnitude is roughly 65–96 basis points per year for a two-standard-deviation increase in log fund size.
- The result survives robustness checks including all funds, 12-month returns, and subperiod splits.

- The negative effect of size is strongest for Small Cap Growth funds and weak for more liquid styles.
- Family size does not hurt performance and may help.
- Small funds and small-cap funds invest more in local stocks and appear to earn better returns on those local investments.
- Multi-manager funds invest less in local stocks and underperform in both local investments and overall fund returns.
- The paper concludes that fund-level scale erodes performance through liquidity and organizational diseconomies.

9 Conclusion

9.1 Core conclusion

Chen, Hong, Huang, and Kubik show that fund size erodes mutual fund performance. The effect is not simply a raw-return pattern; it remains after benchmark adjustment and controls for fund characteristics.

9.2 Economic intuition

A small fund can concentrate capital in its best ideas. A large fund must either take larger positions, which creates liquidity and price-impact costs, or find many more ideas, which requires organizational scale. When the relevant ideas depend on soft information, hierarchy and committee decision-making make this difficult.

9.3 Incremental contribution in one sentence

The paper's incremental contribution is to move the mutual-fund-performance literature from asking only whether managers have skill to asking whether skill is scalable, and to show that scalability depends jointly on market liquidity and fund organization.

9.4 Takeaway

Scale is not always bad, but fund-level scale can be bad when liquidity is limited and the organization cannot generate or process enough high-quality ideas. Family-level scale may help, but the architecture of decision-making matters.